

Arindam Roy Chowdhury

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EDUCATION

Columbia University

Sep 2022 – Jun 2027

Ph.D. in Operations Research (GPA: 4.04/4.00)

Indian Statistical Institute

Jul 2019 – Jun 2021

Master's in Statistics

Indian Statistical Institute

Jul 2016 – May 2019

Bachelor's in Statistics

WORK EXPERIENCE

Susquehanna International Group

Jun 2026 – Aug 2026

Quantitative Researcher Intern

- Developed automated equity trading strategies using signals derived from options-market data.
- Deployed and maintained strategies on continuously released out-of-sample data spanning 2019–2025.
- Ranked in the top quartile among QR interns and new hires based on strategy P&L.

NASA Langley Research Centre

Jul 2024 – Sep 2024

Research Intern

- Formulated tractable optimization framework for modeling multivariate dependency in flight data
- Developed an internal-use Python package; working on MATLAB implementation
- Developed estimation method leveraging non-normalized statistical models

Capital One Financial Services

Jul 2021 – Jul 2022

Associate

- Developed probability-of-default models for real estate and low-default commercial loan portfolios

Research Intern, Data Science Team

May 2020 – Jul 2020

- Developed a procedure for handling missing data in decision-tree models
- Integrated the methodology into an internal Python-based decision-tree tool

Centre for Science of Student Learning

May 2019 – Jul 2019

Research Intern

- Identifying causal impact of different teaching techniques on students

SELECTED PROJECTS

Efficient Uncertainty Quantification of Bagging via the Cheap Bootstrap

Roy Chowdhury, A., Lam, H.

- Developed a lightweight bootstrap framework that yields asymptotically exact confidence intervals for bagged predictors using a small, fixed number of bootstrap resamples with increasing bag size
- Applicable to Random Forests and bagged estimators in Stochastic Optimization

Superiority of Naive Optimization via Stochastic Dominance

Roy Chowdhury, A., Lam, H.

- For stochastic optimization, theoretically established that naive methods like Empirical Risk Minimization (ERM) can outperform approaches such as Distributionally Robust Optimization (DRO) and regularization in minimizing worst-case regret under joint data and distributional uncertainty, leveraging a generalized symmetry argument.
- Demonstrated effectiveness under standard distribution shift regimes widely studied in the literature.

Forecasting Supplies for People in Care

Collaboration with New York City Department of Corrections

- Develop demand forecasting methods for a broad range of goods and supplies required by people in care.

Estimation of Rare Event Probabilities Using Extreme Value Theory

Roy Chowdhury, A., Lam, H.

- Estimating rare-event probabilities by modeling the tail behavior using the Generalized Pareto Distribution (GPD)
- Investigating GPD-based methods as a model-agnostic alternative to importance sampling for rare-event simulation

Bayesian Nonparametric Generalization of Tree Based ML Approaches

Roy Chowdhury, A., Bhattacharya, S.

- Proposed a Bayesian nonparametric mixture model of decision trees with a random number of components, using a Dirichlet Process prior

Accelerography: Feasibility of Gesture Typing using Accelerometer

Roy Chowdhury, A., Dalal, A., Sen, S.

- Designed intuitive motion gestures for the English alphabet, enabling text input by physically moving a phone
- Developed and implemented an R-based pipeline to identify and classify these gestures using accelerometer data.

INVITED TALKS

NASA Langley Research Centre

Nov 2024

Session: *Uncertainty Quantification Seminar*

- **Title 1:** Modelling Multivariate Data using Exponential Polynomials
- **Title 2:** Stochastic Optimization: Superiority of Naive Approaches

INFORMS Annual Meeting

Oct 2024

Session: *Recent Advances in Data-driven Optimization*

- **Title:** Robustness Vs Statistical Efficiency: Superiority of Naive Optimization

The 37th New England Statistics Symposium

May 2024

Session: *Recent Advances in Data-Driven Decision-Making*

- **Title:** Regret Optimality of Empirical Risk Minimization

PATENTS AND PUBLICATIONS

- **Roy Chowdhury, A.**, Crespo, L. G., Lam, H. *Quantification and Decomposition of Uncertainty Using Sliced-Normal Distribution: With Applications to NASA Data*. arXiv:2608.14878, 2026.
- **Roy Chowdhury, A.**, Lam, H. *Efficient Uncertainty Quantification of Bagging via the Cheap Bootstrap*. Proceedings of the 2025 Winter Simulation Conference.
- Saha, P., Dasgupta, J., Tewari, A., Ramesh, A. K. N., **Roy Chowdhury, A.**, S, Anupam., Kedia, G. *Methods and systems for developing decision tree machine learning models*. US 2024/0386287 A1.

AWARDS AND HONOURS

2026	New England Statistics Symposium (NESS): Student Paper Competition Winner
2022	Tang Family Fellowship, Columbia University
2018	Simon Marais Mathematics Competition (top 100 at the undergraduate Asia-Pacific Math competition)
2018	Madhava Mathematics Competition (Ranked within top 25 students at the National level)
2014	Attended the Indian National Math Olympiad Training Camp (INMOTC)
2013	Regional Math Olympiad Awardee (Ranked within top 20 students at the state level)

RELEVANT GRADUATE COURSEWORK

Probability: Probability Theory (Measure Theoretic), Stochastic Modeling

Optimization: Linear Optimization, Discrete Optimization, Convex Optimization

Statistics: Theoretical Statistics, Computational Statistics (Bayesian)

ACADEMIC SERVICES

- Reviewed for Operations Research Letters
- Contributed to editing papers for the Winter Simulation Conference (WSC) 2023